

Report — Analysis of the FIFA World Cup Dataset (1930–2014)

This report summarizes, in plain language, the main findings of the exploratory analysis conducted on 850 World Cup matches, from 1930 to 2014.

1. What Are the Main Patterns Found?

Soccer is a low-scoring sport: most matches end with a total of 1 to 3 goals, and blowouts are rare. Home teams score, on average, almost twice as many goals as visiting teams (1.8 versus 1.0), which confirms the well-known "home-field advantage." Attendance grew steadily between the 1930s and 1980s, stabilizing thereafter — today, stadiums already operate close to their maximum capacity during decisive stages. Knockout-stage matches draw larger crowds than group-stage games, but do not necessarily produce more goals — they are often more cautious and closely contested matches.

2. Which Variables Deserve Greater Attention or Analytical Care?

The attendance column requires care when comparing different stages and decades, as its behavior varies considerably depending on context — it should not be treated as a single, uniform measure. Goals scored at halftime (1st half) are strongly linked to the final score, and therefore should not be used together with the final score in predictive models, due to the risk of double-counting the same information. The column describing the winning condition (Win.conditions) mixes free text with the winning team's name, making it difficult to use directly — it would need to be reorganized into separate fields before any further in-depth analysis.

3. Are There Distinct/Clear Groups of Observations?

Yes. Three groups stand out clearly: (1) group-stage matches versus knockout-stage matches, which have very different attendance profiles; (2) older World Cups (1930-1966), with higher-scoring, more dispersed games, versus more recent World Cups (1998-2014), with more balanced, defensive games; and (3) a small group of historic blowout matches (such as Hungary 10x1 over El Salvador, in 1982) that stand out as genuine extreme events, rather than data-entry errors.

4. Are There Critical Data Quality Issues That Affect the Conclusions?

Yes, two relevant issues were identified. First, 15 matches from the 2014 World Cup are duplicated in the dataset (30 repeated rows), which, if not corrected, artificially inflates the statistics for that edition. Second, even after removing the duplicates, the 2014 World Cup remains one match short of what would be expected for the 32-team format, suggesting a missing record. Both issues were addressed in the analysis by removing the duplicates prior to the calculations, but it is important to document them, as they directly affect any count or average involving the 2014 edition.

5. What Predictive Analyses or Future Studies Could Be Conducted?

The dataset allows, for example: (a) building a predictive model of the match outcome (home win, draw, or away win) using head-to-head history and competition stage; (b) modeling expected attendance as a function of stadium, stage, and teams involved, useful for operational event planning; (c) cross-referencing this dataset with external data on GDP, population, or teams' travel distance, to investigate socioeconomic factors associated with performance; and (d) a more in-depth time-series analysis of the decline in average goals over the decades, testing tactical or regulatory hypotheses (rule changes, athletes' physical evolution).